Department of Electrical Engineering
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Introduction to Prealgebra

Ch. 1 Properties of Arithmetic

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OUTLINE

- Introduction
- Addition
- Multiplication
- Negation
- Subtraction
- Reciprocal
- Division
- Summary

INTRODUCTION

- **What is arithmetic?**
  - The basics of **addition, subtraction, multiplication, division**, and maybe more fancy operations like **square** and **square root**.
  - Mainly directly dealing with numbers. For example

$$2 \times (3 + 5) = (2 \times 3) + (2 \times 5)$$

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ADDITION

- **Example:**
  - How many fruits are there?



3 + 2 =5.



2 + 3 =5.

3 + 2 = 2 + 3.

ADDITION

- **Commutative Property (Commutativity)**

First number + second number = second number + first number

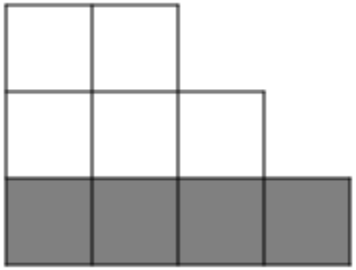
$$a + b = b + a.$$

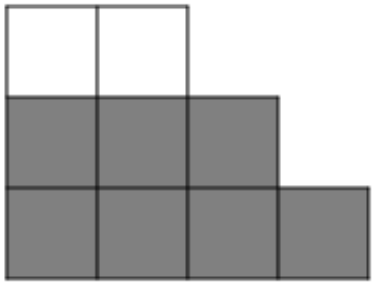
variable representing the first number

variable representing the second number

ADDITION

- **Example**
  - How many squares are in the following pictures?





$$(2+3)+4 = 2 + (3+4).$$

ADDITION

- **Associative Property (Associativity)**
  - Let a, b, and c be numbers. Then

$$(a + b) + c = a + (b + c).$$

ADDITION

- **Example**
  - Compute the sum of 472 + (219 + 28).

ADDITION

- **Example**
  - Compute

$$(2 + 12 + 22 + 32) + (8 + 18 + 28 + 38).$$

ADDITION

- **Example**
  - Compute

$$1 + 2 + 3 + \cdots + 19 + 20.$$

ADDITION

- **Example**
  - Compute

$$50 + 51 + \cdots + 98 + 99 + 100.$$

ADDITION PROPERTIES

- **Commutativity**

$$a + b = b + a.$$

- **Associativity**

$$(a + b) + c = a + (b + c).$$

- **Adding 0**

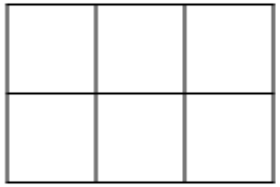
$$a + 0 = a.$$

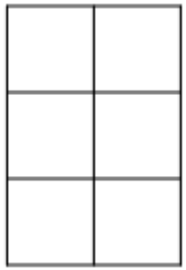
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MULTIPLICATION

- **Example**
  - Use multiplication to find the number of squares in each of the two figures.





$$2 \times 3 = 3 \times 2.$$

MULTIPLICATION

- **Commutative property**
  - Let a and b be numbers. Then

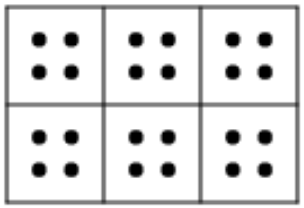
$$a \times b = b \times a.$$

  - Simplified notations of multiplication

$$a \times b = a \cdot b = ab$$

MULTIPLICATION

- **Example**
  - Find the number of dots in the picture below



$$(2 \times 3) \times 4 = 2 \times (3 \times 4).$$

MULTIPLICATION

- **Associative property**
  - Let a, b, and c be numbers. Then

$$(a \times b) \times c = a \times (b \times c).$$

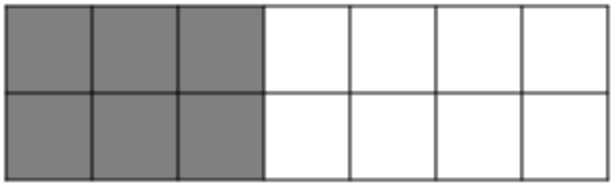
MULTIPLICATION

- **Example**
  - Compute

$$25 \times 125 \times 4 \times 6 \times 8.$$

MULTIPLICATION

- **Example**
  - Using the picture below, explain why $2 \times (3 + 4) = (2 \times 3) + (2 \times 4)$.



MULTIPLICATION

- **Distributive property**

$$a(b+c) = ab + ac.$$

MULTIPLICATION

- **Example**
  - Compute

$$51 \cdot 9 + 51 \cdot 31.$$

MULTIPLICATION

- **Example**
  - What is the value of

$$17 \cdot 13 + 51 \cdot 13 + 32 \cdot 13?$$

MULTIPLICATION

- **Example**
  - What is the value of

$$1 \cdot 100 \cdot 2 \cdot 50 \cdot 4 \cdot 25 \cdot 5 \cdot 20?$$

MULTIPLICATION

- **Example**
  - Compute

$$32 \cdot 16 + 16 \cdot 48.$$

MULTIPLICATION

- **Example**
  - Compute

$$456 + 456 + 456 + 456 + 456 + 456 + 456 + 456 + 456 + 456.$$

MULTIPLICATION

- **Example**
  - What is the product of the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 0?

MULTIPLICATION PROPERTIES

- **Commutativity**

$$a \times b = b \times a.$$

- **Associativity**

$$(a \times b) \times c = a \times (b \times c).$$

- **Distributivity**

$$a(b + c) = ab + ac.$$

- **Multiplication by 1**

$$1 \cdot a = a$$

- **Multiplication by 0**

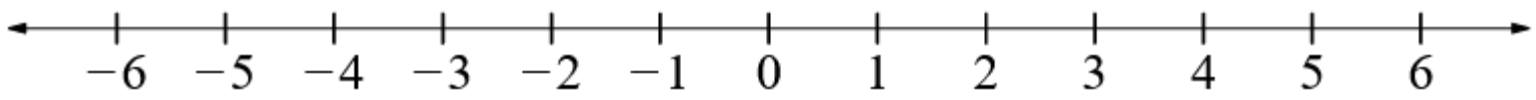
$$0 \cdot a = 0$$

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NEGATION

- **Number line**



$$(-2) + 2 = 0.$$

$$(-6) + 6 = 0.$$

NEGATION

- **Definition**
  - The **negation** of a number *x*, written as –*x*, is the number that we add to *x* to get 0. That is

$$(-x) + x = 0.$$

    - (-*x*) is the **negation** of *x*

  - **Negation** of x is also called as
    - the **opposite** of x, or
    - the **additive inverse** of x.

NEGATION

- **Negation of negation property**

$$-(-x) = x.$$

  - Why?

NEGATION

- **Examples**
  - Find the negation of the following numbers:
    - -1000, -10, -5, 0, 4, 28

NEGATION

- **Multiplying by (-1)**

$$-1 \cdot x = -x.$$

  - Why? (Hint: use the distributivity of multiplication)

NEGATION

- **Multiplying by negation**

$$(-x)y = -(xy).$$
$$x(-y) = -(xy).$$

  - Why? (hint: use multiplying by -1)

NEGATION

- **Negation times negation**

$$(-x)(-y) = xy.$$

  - Why?

NEGATION

- **Negation of sum**

$$-(x+y) = -x - y.$$

  - Why?

NEGATION

- **Example**
  - Compute

$$-631 + (114 + 631).$$

NEGATION

- **Example**
  - What is the sum

$$(-10) + (-9) + (-8) + \cdots + 9 + 10 + 11 + 12.$$

NEGATION

- **Example**
  - What is the value of

$$210 \cdot 5 + 105 \cdot (-9).$$

NEGATION

- **Example**
  - Compute

$$9342 + (-438)719 + (-9340) + (-438)(-719).$$

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SUBTRACTION

- **Definition**
  - Let *a* and *b* be any numbers. Then the **subtraction** *a* – *b* is defined as

$$a - b = a + (-b).$$

  - **The subtraction** *a* – *b* is also called as the **difference** *a* – *b*.

SUBTRACTION

- **Example**

$$2 + 7 = 9.$$
$$2 = 9 - 7.$$
$$7 = 9 - 2.$$

- **Property**

> If $a + b = c$, then $a = c - b$ and $b = c - a$.

  - Why?

SUBTRACTION

- **Example**
  - Calculate the following numbers:

$$11-(-13)=$$
$$-11-13=$$
$$-(13-11)=$$
$$-(11-13)=$$
$$-11+13=$$

- **Some common properties**

$$x-(-y)=x+y$$
$$-x-y=-(x+y)$$
$$-(x-y)=-x+y$$

SUBTRACTION

- **Example**
  - Calculate the following values

$$1643 - 1994 - 1643 =$$

$$268 + 1375 + 6179 - 168 - 1275 - 6079 =$$

SUBTRACTION

- **Example**
  - Calculate the following values

$$85(33 \cdot 22) - 33(22 \cdot 85) =$$

$$3 + (-9) - (-5) =$$

SUBTRACTION

- **Example**
  - If $y - x = 7$, what is $x - y$?

SUBTRACTION

- **Example**
  - Compute

$$(1901 + 1902 + \cdots + 1993) - (101 + 102 + \cdots + 193).$$

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RECIPROCAL

- **Definition**
  - The **reciprocal** of a number *x*, written as $\frac{1}{x}$, is the number such that

$$\frac{1}{x} \cdot x = 1.$$

- **Reciprocal** of *x* is also called as the **multiplicative inverse** of *x*.

- **0 does NOT have a reciprocal.**

RECIPROCAL

- **Reciprocal of reciprocal property**

> Let x be a nonzero number. Then $\frac{1}{x}$ is nonzero and its reciprocal is x.

  - Why?

  - Example
    - What is the reciprocal of $\frac{1}{3}$?

RECIPROCAL

- **Reciprocal of reciprocal property**

  Let x and y be nonzero numbers. Then the reciprocal of xy is

$$\frac{1}{xy} = \frac{1}{x} \cdot \frac{1}{y}.$$

  – Why?

  – Example
    - What is the reciprocal of $\frac{1}{3} \cdot \frac{1}{5} \cdot \frac{1}{7}$?

RECIPROCAL

- **Reciprocal of negation**

  Let x be nonzero numbers. Then the reciprocal of $-x$ is

$$\frac{1}{-x} = -\frac{1}{x}.$$

  – Why?

  – Example
    - What is the reciprocal of $-\frac{1}{7}$?

RECIPROCALS

- **Example**
  - Compute

$$(2 \cdot 3 \cdot 4)\left(\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right).$$

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DIVISION

- **Definition**
  - Let a and b be numbers and b is nonzero. Then the **quotient** $a \div b$ is

$$a \div b = a \times \frac{1}{b}.$$

- **Division by 0 is not defined.**

DIVISION

- **Division property**

> Let a, b, c be nonzero numbers. If $a = b \cdot c$, then
>
> $$a = c \div b,$$
> $$b = c \div a.$$

  - Why?

DIVISION

- **Example**
  - Compute

$$17 \div \frac{1}{5}.$$

DIVISION

- **Division with negation**

$$(-x) \div y = -x \div y,$$
$$x \div (-y) = -(x \div y),$$
$$(-x) \div (-y) = x \div y$$

  - Why?

DIVISION

- **Example**
  - Compute

$$(-10) \div 2,$$
$$10 \div (-2),$$
$$(-10) \div (-2)$$

DIVISION

- **Example**
  - What is

$$20 \cdot 24 \cdot 28 \cdot 32 \div (10 \cdot 12 \cdot 14 \cdot 16)?$$

DIVISION

- **Division property**

$$(a + b) \div c = a \div c + b \div c.$$

– WARNNING:

$a \div (b + c)$ is in general DIFFERENT from $a \div b + a \div c$.

- Example: $12 \div (4 + 2) =$

$12 \div 4 + 12 \div 2 =$

DIVISION

- **Examples**
  - Compute the following numbers

$$(2+4+6+8+10) \div (10+8+6+4+2) =$$

$$28 \div \frac{1}{7} =$$

$$\frac{1}{2} \div \frac{1}{2} \div \frac{1}{2} \div \frac{1}{2} =$$

$$64,000 \div 800 =$$

$$(27 \cdot 31 \cdot 35 \cdot 39 \cdot 43) \div (43 \cdot 39 \cdot 35 \cdot 31) =$$

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SUMMARY

- **Addition**
  - Commutative $a + b = b + a.$
  - Associative $(a + b) + c = a + (b + c).$
  - Adding 0 $a + 0 = a.$

- **Multiplication**
  - Commutative $a \times b = b \times a.$
  - Associative $(a \times b) \times c = a \times (b \times c).$
  - Distributive $a(b + c) = ab + ac.$
  - Multiplying by 1 $1 \cdot a = a.$
  - Multiplying by 0 $0 \cdot a = 0.$

SUMMARY

- **Negation**
  - Definition $-a + a = 0.$
- **Subtraction**
  - Definition $a - b = a + (-b).$
- **Reciprocal**
  - Definition $\frac{1}{a} \cdot a = 1.$
- **Division**
  - Definition $a \div b = a \cdot \frac{1}{b}.$

SUMMARY

- **Order of operations**
  - 1. Evaluate expression inside parenthesis first
  - 2. Compute powers
  - 3. Multiply and divide from left to right
  - 4. Add and subtract from left to right

  - Examples

$$6 \cdot 3 + 4 \cdot (9 \div 3) =$$

$$4 - 8 \cdot (3 + 2) + 4^2 =$$